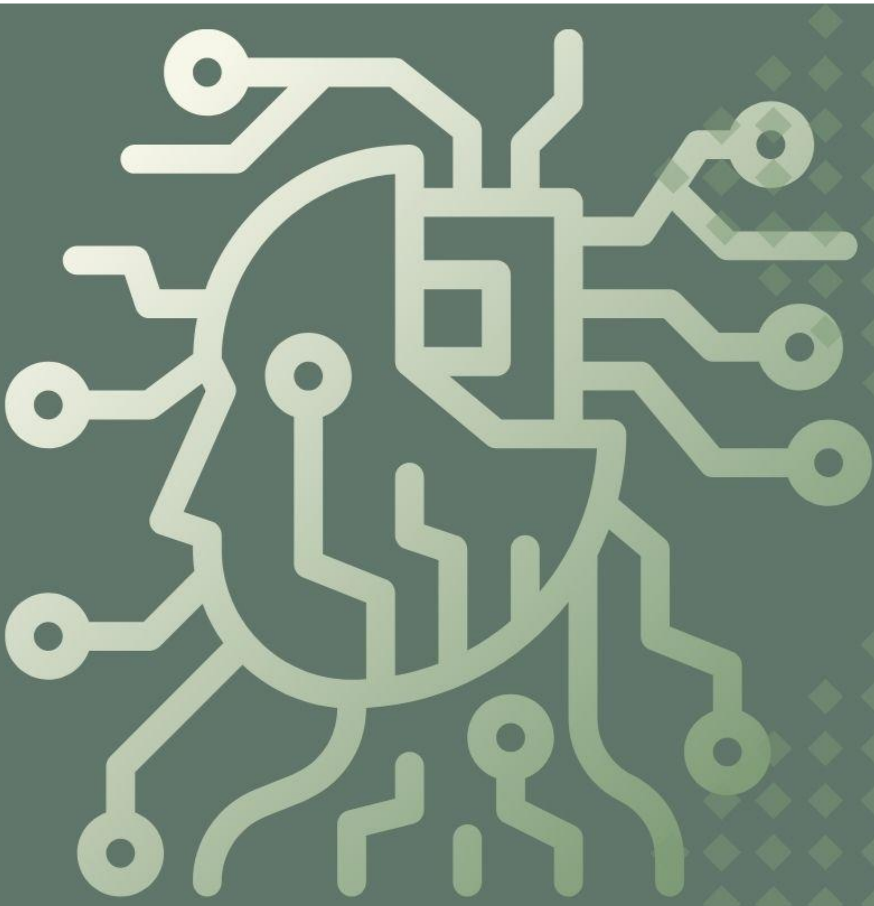

WIRED TO WRITE: TECHNICAL WRITING INSTRUCTIONAL MATERIAL

for Information Technology, Entertainment and Multimedia Computing, Automotive Technology, and Electronics Technology Students

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MODULE 2

Course Outcome:

Utilize effective research and documentation methods.

Lesson 1: Purposes, Audiences, and Characteristics

Specific Learning Objective:

Identify appropriate research methodologies for collecting technical and project-related data in technology fields.

INTO

Activity 1: Get to know

Instruction: Carefully examine the pictures provided, each of which represents a different type of academic resource. For every resource, identify its specific name (for example, journal article, book, government report, database, or research website) and provide a brief explanation of its role in the development of academic papers. Your explanation should highlight how the resource contributes to building credibility, supporting arguments with evidence, or providing background information in scholarly writing.

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THROUGH

Activity 2: Reading

Instructions: Read the provided texts on what technical writing is in the field of biology and environmental science and its purpose, target audience, and characteristics.

Research is a foundation of progress in every academic and professional field. In sociology, community development, and public administration, reliable research is needed to understand human behavior, improve social programs, and strengthen governance. The technology field also benefits from reliable research, as advances in data systems, digital tools, and information management affect the way communities and governments operate.

Methodologies, credible sources, and documentation are interconnected. A researcher cannot rely only on opinions but must use appropriate methods to collect data, evaluate sources that can be trusted, and document the process with clarity. This ensures that the research can be verified, replicated, and applied in solving real life problems.

Research Methodologies for Technical Data

Qualitative research focuses on understanding meanings, experiences, and perspectives. Techniques such as interviews, case studies, and observation are used.

Sociology example: A study on how college students negotiate cultural identity through campus organizations.

Community development example: Conducting focus group discussions with residents to understand their views on water supply issues.

Public administration example: Observing how citizens interact with local government offices to identify barriers in service delivery.

Quantitative research deals with numerical data and measurable outcomes. Methods include experiments, surveys, and statistical data analysis.

Sociology example: Using a survey to measure the relationship between social media use and student academic performance.

Community development example: Collecting data on the number of families benefiting from livelihood programs.

Public administration example: Analyzing government performance reports to evaluate efficiency in public spending.

Mixed methods combine qualitative and quantitative techniques. This is common in applied fields where both numbers and lived experiences are essential. Approaches include design projects, prototypes, and user testing.

Sociology example: Designing a project that uses both survey data and life stories to understand the impact of unemployment on young adults.

Community development example: Developing a prototype community education program and testing it with residents while also gathering statistical feedback.

Public administration example: Creating a citizen feedback system, first tested through focus groups, then assessed using survey data to measure satisfaction.

Activity 3: Data Mapping

Instructions: Students will be divided into groups according to their field of study. Each group will be assigned a research scenario that requires the collection of project-related data in the technology field. The tasks for each discipline are as follows:

Sociology: Students will design a short online survey using applications such as Google Forms. The survey should focus on measuring how students within their campus use digital platforms for communication.

Community Development: Students will use GIS mapping tools, such as Google My Maps, to plot the availability of digital health kiosks or ICT hubs in nearby communities.

They will analyze the data to identify access patterns and possible gaps in service delivery.

Public Administration: Students will utilize social media analytics tools, such as Facebook Insights or Twitter Analytics, to collect engagement data from public information campaigns conducted by local agencies or organizations.

Output

After completing the data collection process within the allotted time, each group will organize their findings into a brief presentation. The presentation should explain how the chosen tool enabled them to gather reliable and project-relevant information.

Discussion and Reflection

The activity will conclude with a class sharing session. Each group will present their scenario, the tool they used, and their findings. They should also highlight both the strengths and limitations of the methodology in addressing their assigned project. A guided discussion will follow, allowing students to reflect on the effectiveness of the different methodologies and the importance of selecting appropriate tools in technology-driven research.

BEYOND

Activity 4: What are your thoughts?

Instructions: After completing the group activity and presentations, each student is required to write an individual reflection. In the reflection, students should describe what they learned about the use of digital tools survey applications, GIS mapping, or social media analytics in collecting project-related data. They should connect the experience to their field of study (sociology, community development, or public administration) by discussing how the activity helped them in getting the data.

Essay rubric

Criteria	Excellent (3)	Satisfactory (2)	Needs Improvement (1)
Depth of Insight	Demonstrates strong understanding and critical analysis of the methodology	Shows general understanding with some analysis	Shows limited or surface-level understanding
Connection to Field of Study	Clearly connects learning to sociology, community development, or public administration	Some connection to field, but not fully developed	Weak or missing connection to field
Evaluation of Method	Provides balanced strengths and weaknesses with justification	Mentions either strengths or weaknesses with limited detail	Little or no evaluation of the methodology
Creativity in Improvement	Suggests thoughtful and innovative ideas for improvement or combining methods	Suggests some improvements, but not fully explained	No clear suggestions for improvement
Clarity and Organization	Reflection is well-organized, clear, and within word count	Reflection is understandable but may lack clarity or organization	Reflection is unclear, disorganized, or incomplete